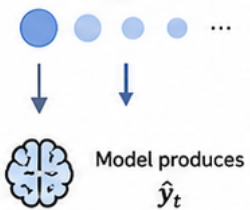


1. Streaming Input

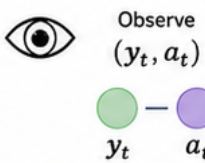
Data stream of triplets

$$(x_t, y_t, a_t)$$

(x_t) arrives



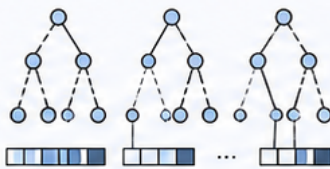
$(\hat{y}_t)$ prediction



**Prequential
(test-then-train)**

2. Fair Oblique Forest

soft-routed trees;
fairness penalty in updates



$$f_t(x) = \frac{1}{M} \sum_{m=1}^M n_m(x)$$

Node-level
stats
 $n_{ij}(x)$

Updated
forest
parameters

7. Fairness Monitor

Rolling window (size W)

Compute fairness metrics

- Demographic Parity (DP)
- Equalized Odds (EO)
- Equal Opportunity (TPR parity)

...

Metric
curves

3. Performance Monitor

Rolling window (size W)

Error stream input



Metrics (rolling window)

- Accuracy
- Demographic Parity (DP)
- Equalized Odds (EO)
- ...

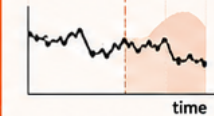
Prediction
error stream
 $e_t = 1[\hat{y}_t \neq y_t]$



$\hat{y}_t$

e_t

4. ADWIN Drift Detector



δ_{warn} (warning threshold)

δ_{confirm} (drift threshold)

Warning

Drift
confirmed

5. Reaction Controller

Modify training hyperparameters



Warning:

- Increase learning rate ($\times 2$)
- Keep routing temperature (τ) high



Confirmed drift:

- Larger learning rate spike ($\times 4$)
- Reduce routing temperature (τ)
- Reset / refresh affected trees (if applicable)



Recovery:

- Anneal learning rate and τ back to baseline
- Cooldown window before next adaptation

6. Training Update

Test-then-train update

(x_t, y_t, a_t)

Fairness penalty
(node-level)

Controller actions
(LR, τ , reset flags)

Update forest
parameters
(gradient step)

Adam optimizer

8. Logging / Visualization

Time series under drift phases



Accuracy over time



DP / EO over time

Drift Phases



Legend

Model / data flow (predictions, learning, metrics)

Drift signals and control flow (detection, reaction)

All metrics are computed over a rolling window of the most recent W samples.
The system operates online with batch size 1 under a prequential (test-then-train) protocol.